

MAKING PCBs AT HOME

Photographically



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Part 1 of this article in previous issue explained creation of schematic and PCB layout. This part covers the remaining steps for making PCBs photographically at home, starting with preparation of the artwork.

Step 3: Prepare the artwork

Before you prepare the artwork for transferring the layout to the pre-sensitised PCB laminate, it will be worthwhile understanding how the technique for making PCBs photographically works. For positive-acting type boards, you need a positive artwork film (see Fig. 21), which means that areas where you want to retain copper would be black and opaque to UV light. The UV light passes through the artwork and strikes the material where you want to remove copper, chemically changing the photographic material coated on the PCB such that it becomes soluble in the photoresist developer solvent. This exposes the underlying copper, which is removed by the etching process as described later.

Use A4-size transparencies that were traditionally used for preparing OHP presentation slides. These are readily available at most stationery shops for around ₹ 5 apiece. Ideally, you should use transparencies specially made for laser printers, but these are not so readily available and cost ₹ 50 to ₹ 100 per piece.

Print the artwork through a laser printer. Use the setting for the darkest printout on a black-and-white laser printer. The best results are, however, achieved by using a colour laser printer set to print in colour mode at the highest quality setting. This ensures that multiple layers of colour ink get deposited on the transparency, making the dark areas of the artwork absolutely opaque.

Just to reiterate, the final success largely depends on the quality of artwork you prepare. This means that you need a clear

and sharp image for the artwork, which has black areas solid black and opaque to UV light.

Tips, pitfalls and workarounds. Before taking the final printout on a transparency, take a printout on a plain paper and place some components on it to ensure that there is no scaling done either inadvertently or by the software. This is a vital step as many people end up making great looking PCBs only to realise that component pads do not match pins of the actual components.

Many printers produce printouts that are really not dark enough and do not have enough ink deposited on the transparency to make black areas opaque to even ordinary light, leave alone UV light. I usually take four copies of printed transparencies, align them one on top of another and then glue them together on the area around the artwork. This provides excellent results. You must invest time to carefully inspect your artwork.

Create two stacks of books and place them about 20cm (8 inches) apart (see Fig. 22). Place a rectangular piece of transparent glass on top of the two stacks so that it bridges them. Place a small LED or CFL bulb below the glass. Place a transparency on top of the glass and inspect it, preferably with a magnifying glass. Use a permanent marking pen with a small tip to correct any black area that is not dark or opaque enough. When done, stack up the second transparency on top of the first one and carefully align them. Then use fast-setting glue to stick the two aligned transparencies outside of the artwork area. Repeat the process for the other two transparencies. Now you should have a sharp and opaque artwork.

Step 4: Transfer the artwork

Before you transfer the artwork onto the PCB laminate using a UV light source, cut